Math 240 Matlab Project 1

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*Type/paste Matlab commands for a particular problem/part below the correspondingly labeled line, above the next double % sign. Written responses to questions (not lines of code) should be given as comments, following a single percent symbol % NEWLINE

Problem 0

No

Problem 1

(a)

```
B = [2 3 1 10; -6 -2 -5 -36; 2 17 2 -57]
```

```
B =

     2     3     1    10
    -6    -2    -5   -36
     2    17     2   -57
```

(b)

```
B(3,:) = B(3,:) - B(1,:)
B(2,:) = B(2,:) + 3 * B(1,:)
B(3,:) = B(3,:) - 2 * B(2,:)
B(3,:) = 1/5 * B(3,:)
```

```
B =

     2     3     1    10
    -6    -2    -5   -36
```

```
     0    14     1   -67
```

```
B =

     2     3     1    10
     0     7    -2    -6
     0    14     1   -67
```

```
B =

     2     3     1    10
     0     7    -2    -6
     0     0     5   -55
```

```
B =

     2     3     1    10
     0     7    -2    -6
     0     0     1   -11
```

(c)

```
B(1,:) = B(1,:) - B(3,:)
B(2,:) = B(2,:) + 2 * B(3,:)
B(2,:) = 1/7 * B(2,:)
B(1,:) = B(1,:) - 3 * B(2,:)
B(1,:) = 1/2 * B(1,:)
```

```
B =

     2     3     0    21
     0     7    -2    -6
     0     0     1   -11
```

```
B =

     2     3     0    21
     0     7     0   -28
     0     0     1   -11
```

```
B =

     2     3     0    21
     0     1     0    -4
     0     0     1   -11
```

```
B =
```

```
     2     0     0    33
     0     1     0    -4
     0     0     1   -11


B =

    1.0000         0         0   16.5000
         0    1.0000         0   -4.0000
         0         0    1.0000  -11.0000
```

(d) x1 = 16.5 x2 = -4 x3 = -11

Problem 2

(a)

```
B=[1 -4 9 -10 2 13;3 -10 27 -31 13 35;1 -4 9 -8 -8 17;-2 18 -18 7 71 -62]


B =

     1    -4     9   -10     2    13
     3   -10    27   -31    13    35
     1    -4     9    -8    -8    17
    -2    18   -18     7    71   -62
```

(b)

```
B(4,:) = B(4,:) + 2 * B(1,:)
B(3,:) = B(3,:) - B(1,:)
B(2,:) = B(2,:) - 3 * B(1,:)
B(4,:) = B(4,:) - 5 * B(2,:)
B(4,:) = B(4,:) + 4 * B(3,:)
B(3,:) = 1/2 * B(3,:)


B =

     1    -4     9   -10     2    13
     3   -10    27   -31    13    35
     1    -4     9    -8    -8    17
     0    10     0   -13    75   -36


B =

     1    -4     9   -10     2    13
     3   -10    27   -31    13    35
     0     0     0     2   -10     4
     0    10     0   -13    75   -36
```

```
B =

     1    -4     9   -10     2    13
     0     2     0    -1     7    -4
     0     0     0     2   -10     4
     0    10     0   -13    75   -36

B =

     1    -4     9   -10     2    13
     0     2     0    -1     7    -4
     0     0     0     2   -10     4
     0     0     0    -8    40   -16

B =

     1    -4     9   -10     2    13
     0     2     0    -1     7    -4
     0     0     0     2   -10     4
     0     0     0     0     0     0

B =

     1    -4     9   -10     2    13
     0     2     0    -1     7    -4
     0     0     0     1    -5     2
     0     0     0     0     0     0
```

(c)

```
B(2,:) = B(2,:) + B(3,:)
B(1,:) = B(1,:) + 10 * B(3,:)
B(1,:) = B(1,:) + 2 * B(2,:)
B(2,:) = 1/2 * B(2,:)
```

```
B =

     1    -4     9   -10     2    13
     0     2     0     0     2    -2
     0     0     0     1    -5     2
     0     0     0     0     0     0

B =

     1    -4     9     0   -48    33
     0     2     0     0     2    -2
     0     0     0     1    -5     2
```

```
     0     0     0     0     0     0
```

```
B =

     1     0     9     0   -44    29
     0     2     0     0     2    -2
     0     0     0     1    -5     2
     0     0     0     0     0     0
```

```
B =

     1     0     9     0   -44    29
     0     1     0     0     1    -1
     0     0     0     1    -5     2
     0     0     0     0     0     0
```

(d)

```
B=[1 -4 9 -10 2 13;3 -10 27 -31 13 35;1 -4 9 -8 -8 17;-2 18 -18 7 71 -62]
rref(B)
```

```
B =

     1    -4     9   -10     2    13
     3   -10    27   -31    13    35
     1    -4     9    -8    -8    17
    -2    18   -18     7    71   -62
```

```
ans =

     1     0     9     0   -44    29
     0     1     0     0     1    -1
     0     0     0     1    -5     2
     0     0     0     0     0     0
```

(e)

```
%[x1;x2;x3;x4;x5] = [29;-1;0;2;0] + x3*[-9;0;1;0;0] + x5*[44;-1;0;-5;1]
```

Problem 3

(a)

```
format short
A = [2 -4 -1 4; 5 4 7 8; -7 -2 -5 5]
rref(A)
```

```
A =
```

```
     2    -4    -1     4
     5     4     7     8
    -7    -2    -5     5


ans =

    1.0000         0         0   -4.3636
         0    1.0000         0   -4.9545
         0         0    1.0000    7.0909
```

(b) x1 = -4.3636 x2 = -4.9545 x3 = 7.0909

(c)

```
format rat
rref(A)


ans =

       1             0             0           -48/11
       0             1             0          -109/22
       0             0             1            78/11
```

(d) x1 = -48/11 x2 = -109/22 x3 = 78/11

Problem 4

(a)

```
format short
% 270x1 + 51x2 + 70x3 = 400
% 10x1 + 5.4x2 + 15x3 = 30
% 2x1 + 5.2x2 + 0x3 = 10
A = [270 51 70 400; 10 5.4 15 30; 2 5.2 0 10]
rref(A)
% 0.9858 servings of Mac & Cheese
% 1.5439 servings of Broccoli
% 0.7870 servings of Chicken


A =

  270.0000   51.0000   70.0000  400.0000
   10.0000    5.4000   15.0000   30.0000
    2.0000    5.2000         0   10.0000


ans =
```

```
    1.0000         0         0    0.9858
         0    1.0000         0    1.5439
         0         0    1.0000    0.7870
```

(b) 260x1 + 51x2 + 70x3 = 400 9x1 + 5.4x2 + 15x3 = 30 5x1 + 5.2x2 + 0x3 = 10

```
A = [260 51 70 400; 9 5.4 15 30; 5 5.2 0 10]
rref(A)
% 1.0890 servings of Shells
% 0.8760 servings of Broccoli
% 1.0313 servings of Chicken


A =

  260.0000   51.0000   70.0000  400.0000
    9.0000    5.4000   15.0000   30.0000
    5.0000    5.2000         0   10.0000


ans =

    1.0000         0         0    1.0890
         0    1.0000         0    0.8760
         0         0    1.0000    1.0313
```

Problem 5

(a)

```
A = [14 -16 -2; 50 10 10; 39 -11 3]
rref(A)


A =

    14   -16    -2
    50    10    10
    39   -11     3


ans =

    1.0000         0    0.1489
         0    1.0000    0.2553
         0         0         0
```

(b) Yes, the given vector is in the span of the other two. Since the linear system's augmented matrix is consistent.

(c) The 3 vectors are linearly dependent because the last row is made up of all 0s in rref. This means that one of the vectors can be written as a linear combination of the other 2.

Problem 6

(a)

```
syms a
syms b
```

(b)

```
A = [5 9 a; 8 -6 b]
rref(A)
```

```
A =

[5,  9, a]
[8, -6, b]


ans =

[1, 0,      a/17 + (3*b)/34]
[0, 1, (4*a)/51 - (5*b)/102]
```

(c) w1 = a/17 + (3*b)/34 w2 = (4*a)/51 - (5*b)/102

Problem 7

(a)

```
A = [2 -2 -4 2 -1; 1 8 3 -3 2; -6 -5 -6 -13 -9; 9 -2 -9 11 8]
rref(A)
```

```
A =

     2    -2    -4     2    -1
     1     8     3    -3     2
    -6    -5    -6   -13    -9
     9    -2    -9    11     8


ans =

     1     0     0     2     0
     0     1     0    -1     0
     0     0     1     1     0
     0     0     0     0     1
```

(b) There are only 4 pivots even though there are 5 vectors. This means that one of the vectors can be written as a linear combination of the others showing that these set of vectors are linearly dependent.

(c)

```
A = [2 -2 -4 2 -1 0; 1 8 3 -3 2 0; -6 -5 -6 -13 -9 0; 9 -2 -9 11 8 0]
rref(A)
% -2v1 + v2 + -v3 + 0v5 = 0
```

```
A =

     2    -2    -4     2    -1     0
     1     8     3    -3     2     0
    -6    -5    -6   -13    -9     0
     9    -2    -9    11     8     0


ans =

     1     0     0     2     0     0
     0     1     0    -1     0     0
     0     0     1     1     0     0
     0     0     0     0     1     0
```

(d) Theorem 8: If a set contains more vectors than there are entries in each vector, then the set is linearly dependent.

(e) Yes, this set of vectors does span R^4 because there are 4 pivots which means that there are at least 4 vectors that are linearly independent and cannot be written as a linear combination of the other vectors. This means there are 4 dimensions.

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